

Number Theory and Proof Methods

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4.1 Introduction

4.2 Rational Numbers

4.3 Divisibility

4.4 Quotient-Remainder Theorem



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Acknowledgement:

This lecture is based on, but not limited to, chapter 4 in “Discrete Mathematics with Applications by Susanna S. Epp (3rd Edition)”.

Number Theory

4.4 Quotient-Remainder Theorem

In this lecture:



☐ Part 1: **Quotient-Remainder Theorem**

☐ Part 2: *div* and *mod*, and applications in real-life

☐ Part 3: Representing Integers in Quotient-Remainder

☐ Part 4: Absolute Value

Keywords: Number Theory, Quotient-Remainder Theorem, *div*, *mod*, divide into cases” Proof Method, Parity, Integers Modulo, Absolute Value

Quotient-Remainder Theorem

Notice that:

$$\begin{array}{r}
 2 \leftarrow \text{quotient} \\
 4 \overline{) 11} \\
 \underline{8} \\
 3 \leftarrow \text{remainder}
 \end{array}
 \qquad
 11 = 2 \cdot 4 + 3.$$

$\uparrow$ $\uparrow$
 2 groups of 4 3 left over

Theorem 4.4.1 The Quotient-Remainder Theorem

Given any integer n and positive integer d , there exist unique integers q and r such that

$$n = dq + r \quad \text{and} \quad 0 \leq r < d.$$

Examples:

$$54 = 4 \cdot 13 + 2$$

$$q = 13 \quad r = 2$$

$$-54 = 4 \cdot (-14) + 2$$

$$q = -14 \quad r = 2$$

$$54 = 70 \cdot 0 + 54$$

$$q = 0 \quad r = 54$$

Number Theory

4.4 Quotient-Remainder Theorem

In this lecture:

- ☐ Part 1: Quotient-Remainder Theorem
-  ☐ Part 2: ***div* and *mod*, and applications in real-life**
- ☐ Part 3: Representing Integers in Quotient-Remainder
- ☐ Part 4: Absolute Value

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div and mod

• Definition

Given an integer n and a positive integer d ,

$n \text{ div } d$ = the integer quotient obtained
when n is divided by d , and

$n \text{ mod } d$ = the nonnegative integer remainder obtained
when n is divided by d .

Symbolically, if n and d are integers and $d > 0$, then

$$n \text{ div } d = q \quad \text{and} \quad n \text{ mod } d = r \quad \Leftrightarrow \quad n = dq + r$$

where q and r are integers and $0 \leq r < d$.

"/" in C++, JAVA, .net

"%" in C, JAVA
"\n" in .net

Examples:

$$32 \text{ div } 9 = 3$$

$$32 \text{ mod } 9 = 5$$

Application of div and mod

Computing the Day of the Week

Suppose today is Tuesday, and neither this year nor next year is a leap year (سنة كبيسة). What day of the week will it be 1 year from today?

$$365 \text{ div } 7 = 52 \quad \text{and} \quad 365 \text{ mod } 7 = 1$$

So,
after 364 it will be Tuesday, and after 365 it will be Wednesday

Application of div and mod

Computing the Day of the Week

If today is Saturday and it is 16/10/2021, which day it will be on 20/2/2022?

The number of days from today to 20/2/2022 = 15 in October + 30 in November + 31 in December + 31 in January + 20 in February = 127 days

$$127 \text{ div } 7 = 18 \quad 127 \text{ mod } 7 = 1$$

That is, after 18 weeks the day will be Saturday, and one day after, it will be Sunday

Application of div and mod

Solving a Problem about *mod*

Suppose m is an integer. If $m \bmod 11 = 6$,
what is $4m \bmod 11$?

$$m = 11q + 6$$

$$\begin{aligned}\text{So, } 4m &= 44q + 24 \\ &= 44q + 22 + 2 \\ &= 11(\underline{4q + 2}) + 2\end{aligned}$$

$(4q + 2)$ is integer

$$\text{Thus } 4m \bmod 11 = 2$$

Number Theory

4.4 Quotient-Remainder Theorem

In this lecture:

- ☐ Part 1: Quotient-Remainder Theorem
- ☐ Part 2: *div* and *mod*, and applications in real-life
-  ☐ Part 3: **Representing Integers in Quotient-Remainder**
- ☐ Part 4: Absolute Value

Keywords: Number Theory, Quotient-Remainder Theorem, *div*, *mod*, divide into cases” Proof Method, Parity, Integers Modulo, Absolute Value

Representing Integers using the quotient-remainder theorem

Parity Property

We represent any number as:

$$n = 2q + r \quad \text{and} \quad 0 \leq r < 2$$

Because we have only $r = 0$ and $r = 1$, then:

$$\begin{array}{ccc} n = 2q + 0 & \text{or} & n = 2q + 1 \\ \text{Even} & & \text{Odd} \end{array}$$

Therefore, n is either even or odd (parity)

Representing Integers using the quotient-remainder theorem

Proving Parity Property

Theorem 4.4.2 The Parity Property

Any two consecutive integers have opposite parity.

Proof:

Given m and $m+1$ are consecutive integers

Then, one is odd and the other is even (by parity property)

Case1 (m is even): $m = 2k$, so $m + 1 = 2k + 1$, which is odd

Case2 (m is odd): $m = 2k + 1$ and so $m+1 = (2k+1) + 1 = 2k + 2 = 2(k+1)$.

thus $m + 1$ is even.

Proof by division into cases

The “divide into cases” Proof Method

Method of Proof by Division into Cases

To prove a statement of the form “If A_1 or A_2 or $\dots$ or A_n , then C ,” prove all of the following:

If A_1 , then C ,

If A_2 , then C ,

$\vdots$

If A_n , then C .

This process shows that C is true regardless of which of $A_1, A_2, \dots, A_n$ happens to be the case.

Representing Integers using the quotient-remainder theorem

Integers Modulo 4

We represent any integer as:

$$n=4q \quad \text{or} \quad n=4q+1 \quad \text{or} \quad n=4q+2 \quad \text{or} \quad n=4q+3$$

This implies that there exist an integer quotient q and a remainder r such that

$$n = 4q + r \quad \text{and} \quad 0 \leq r < 4.$$

Using the “divide into cases” Proof Method

Theorem 4.4.3

The square of any odd integer has the form $8m + 1$ for some integer m .

Proof: $\forall n \in \text{Odd}, \exists m \in \mathbb{Z} . n^2 = 8m + 1.$

Hint: any odd integer can be $(4q+1)$ or $(4q+3)$.

Case 1 ($n=4q+1$):

$$n^2 = 8m + 1 = (4q+1)^2 = 16q^2 + 8q + 1 = 8(\underline{2q^2 + q}) + 1$$

$(2q^2 + q)$ can be is an integer m , thus $n^2 = 8m + 1$

Case 2 ($4q+3$):

$$n^2 = 8m + 1 = (4q+3)^2 = 16q^2 + 24q + 8 + 1 = 8(\underline{2q^2 + 3q+1}) + 1$$

$(2q^2 + 3q+1)$ can be is an integer m , thus $n^2 = 8m + 1$

Number Theory

4.4 Quotient-Remainder Theorem

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- ❑ Part 1: Quotient-Remainder Theorem
- ❑ Part 2: *div* and *mod*, and applications in real-life
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 ❑ Part 4: **Absolute Value**

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Absolute Value

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Definition

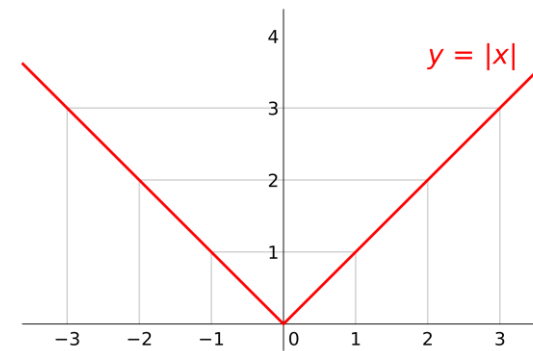
For any real number x , the **absolute value of x** , denoted $|x|$, is defined as follows:

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

Example:

$$|2| = 2$$

$$|-2| = 2$$



Absolute Value

Lemma 4.4.4

For all real numbers r , $-|r| \leq r \leq |r|$.

Proof:

Suppose r is any real number. We divide into cases according to whether $r \geq 0$ or $r < 0$.

Case 1 ($r \geq 0$): by definition $|r| = r$. Also, r is positive and $-|r|$ is negative, $\Rightarrow -|r| < r$.

Case 2 ($r < 0$): by definition $|r| = -r$, thus, $-|r| = r$. Also r is negative and $|r|$ is positive. $\Rightarrow r < |r|$.

Thus, in either case, $-|r| \leq r \leq |r|$

Absolute Value

Lemma 4.4.5

For all real numbers r , $|-r| = |r|$.

T4. $-(-a) = a$.

T23. If $a < b$ and $c < 0$, then $ac > bc$.

T24. If $a < b$, then $-a > -b$. In particular, if $a < 0$, then $-a > 0$.

Proof: Suppose r is any real number. By Theorem T23 in Appendix A, if $r > 0$, then $-r < 0$, and if $r < 0$, then $-r > 0$. Thus

$$\begin{aligned}
 |-r| &= \begin{cases} -r & \text{if } -r > 0 \\ 0 & \text{if } -r = 0 \\ -(-r) & \text{if } -r < 0 \end{cases} && \text{by definition of absolute value} \\
 &= \begin{cases} -r & \text{if } -r > 0 \\ 0 & \text{if } -r = 0 \\ r & \text{if } -r < 0 \end{cases} && \begin{array}{l} \text{because } -(-r) = r \text{ by Theorem T4} \\ \text{in Appendix A} \end{array} \\
 &= \begin{cases} -r & \text{if } r < 0 \\ 0 & \text{if } -r = 0 \\ r & \text{if } r > 0 \end{cases} && \begin{array}{l} \text{because, by Theorem T24 in Appendix A, when} \\ -r > 0, \text{ then } r < 0, \text{ when } -r < 0, \text{ then } r > 0, \\ \text{and when } -r = 0, \text{ then } r = 0 \end{array} \\
 &= \begin{cases} r & \text{if } r \geq 0 \\ -r & \text{if } r < 0 \end{cases} && \text{by reformatting the previous result} \\
 &= |r| && \text{by definition of absolute value.}
 \end{aligned}$$

Absolute Value and Triangle Inequality

Theorem 4.4.6 The Triangle Inequality

For all real numbers x and y , $|x + y| \leq |x| + |y|$.

Proof:

Case 1 ($x + y \geq 0$): $|x + y| = x + y$ by Lemma 4.4.4,
and so, $x \leq |x|$ and $y \leq |y|$
hence, $|x + y| = x + y \leq |x| + |y|$

Case 2 ($x + y < 0$): $|x + y| = -(x + y) = (-x) + (-y)$ by Lemmas 4.4.4 & 4.4.5
and so, $-x \leq |-x| = |x|$ and $-y \leq |-y| = |y|$.
hence, $|x + y| = (-x) + (-y) \leq |x| + |y|$.